Bogleheads

Asset allocation

*This article contains details **specific to United States (US) investors**. Parts of it are not intended for non-US investors.*
Non-US investors can find related information at Stock asset allocation for non-US investors and Building a non-US Boglehead portfolio.

Asset allocation means dividing an investment portfolio among different asset classes. Typically these are stocks, bonds, and cash.[1][note 1] Determining which mix of assets to hold in a portfolio is a personal choice. The asset allocation that works best at any given stage in your life will depend largely on your need, ability and willingness take risk. These depend on the investment *time horizon*, and on your *financial capacity* and *emotional capacity* to tolerate risk and to stay the course.

While this may sound like a daunting task, there are straightforward guidelines to help you choose an appropriate asset allocation.

How much in bonds? How much in stocks? That is the basic question of asset allocation. The more risk you can handle, the less bonds you need. When you are young, your prime earning years lie ahead, and it will be decades before you need to access the money. So higher stock allocations may be suitable, because big drops in stock prices will not hurt as long as you do not flee the market.

John Bogle said that "as we age, we usually have (1) more wealth to protect, (2) less time to recoup severe losses, (3) greater need for income, and (4) perhaps an increased nervousness as markets jump around. All four of these factors suggest more bonds as we age."[2]

Your asset allocation applies to your overall portfolio. You do not need to have the same asset allocation in every account you own; in fact, this may be detrimental. See Asset allocation in multiple accounts and Tax-efficient fund placement for more details.

Strategic asset allocation

A key reason for having an asset allocation strategy is to help you reduce the risks of volatile equity asset classes, ones that are expected to provide higher returns, by combining them with more stable fixed-income assets. These balanced portfolios help reduce volatility and down-side risk, making it easier for you to maintain a long term investment program (stay the course) without panic selling during bear markets.

With a **strategic asset allocation** - as explained in this article - you choose a base target allocation to a selection of different asset classes for the long run. Strategic asset allocation is used by investors following the Bogleheads® investment philosophy. You can change it due to life events, but you should not change it because of market conditions. The main asset classes are equities (stock), fixed-income (bonds) and cash.

The goal is to select an asset allocation that lets you sleep at night, and *avoid* the destructive urge to sell out in a panic the next time the market plummets; then having to agonize over when it is a "good time' to get back in. This leads to selling low and buying high, the exact opposite of prudent investing.

Because the three main asset classes - equities, fixed-income, and cash equivalents - have different levels of risk and return, each will behave differently over time. Because of this, you will need to periodically rebalance the allocation to these asset classes back to their target allocation to keep the portfolio's risk-return level constant.[3]

Alternatively, a **tactical asset allocation** shifts allocations according to economic or valuation factors. Vanguard has historically used tactical asset allocation for a limited number of its balanced funds. Finally, a **dynamic asset allocation** shifts allocations in accordance with changing future liabilities.

Stocks and bonds

Main articles: Stock basics and Bond basics

You need stocks to get the expected return needed to accumulate funds for retirement. Stocks provide investors with a share of the profits generated by publicly owned companies in the economy. But in exchange for the hope of high return, stocks are extremely volatile and risky. Many investors learned how risky stocks can be in 2008 when they fell 50% from their previous highs. Over time, stock prices roughly follow the trend of the economy, which is to grow. But prices can stagnate or decline for decade-long periods. This is why an allocation to bonds is a necessary part of an asset allocation.[4]

Bonds are a promise to pay back a loan of money on a pre-set schedule. Bonds do not produce the same expected high returns that stocks do, but they are much less volatile. The way to get reasonable growth without stomach-churning drops is to hold a mix of stocks and bonds.

Impact of asset allocation on risk and return

> In short, during the next 20 or 30 years, there will be a single, best allocation that in retrospect we will have wished we have owned. The only problem is that we haven't a clue what that portfolio will be. So, the safest course is to own as many asset classes as you can; that way you can be sure of avoiding the catastrophe of holding a portfolio concentrated in the worst ones.
>
> — William J. Bernstein, The Four Pillars of Investing, pg 244, McGraw Hill

The first table below shows the returns of various stock/bond allocations from 2000 - 2002. This period was marked by falling stock prices. The second table reflects the longer-term rewards investors hope to receive, assuming that the historical pattern of bond returns providing a premium return over inflation, and stock returns providing a premium over bond returns will be realized. The tables show why asset allocation is important. It determines future return, as well as the bear market burden of periodic losses that you will have to carry successfully to get those returns.

Cumulative return after inflation from 2000-to-2002 bear market[5]	
80% stock / 20% bond	-34.35%
70% stock / 30% bond	-25.81%
60% stock / 40% bond	-19.99%
50% stock / 50% bond	-13.87%
40% stock / 60% bond	-7.46%
30% stock / 70% bond	-0.74%
20% stock / 80% bond	+6.29%

Projected 10 year Cumulative return after inflation stock return 8% yearly, bond return 4.5% yearly, inflation 3% yearly[6]	
80% stock / 20% bond	52%
70% stock / 30% bond	47%
60% stock / 40% bond	42%
50% stock / 50% bond	38%
40% stock / 60% bond	33%
30% stock / 70% bond	29%
20% stock / 80% bond	24%

Rules of thumb

Although your exact asset allocation should depend on your goals for the money, there are some *rules of thumb* to help.[note 2] Any rule of thumb is only a starting point for decision making, not the end.

The most important asset allocation decision is the split between risky and non-risky assets. This is most often referred to as the stock/bond split.

Benjamin Graham's timeless advice was:[7][note 3]

> We have suggested as a fundamental guiding rule that the investor should never have less than 25% or more than 75% of his funds in common stocks, with a consequence inverse range of 75% to 25% in bonds. There is an implication here that the standard division should be an equal one, or 50-50, between the two major investment mediums.
>
> — Ben Graham, The Intelligent Investor

John Bogle recommended "roughly your age in bonds"; for instance, at age 45, about 45% of the portfolio should be allocated to high-quality bonds. He also suggested that, during the retirement distribution phase, you should treat the value of any future pension and expected Social Security payments as a bond-like component of wealth and asset allocation.[note 4]

If you choose less conservative guidelines, you should understand why you feel that you have the *need, ability, and willingness* to take on the greater inherent risk, as explained in the next section.

All these age-based guidelines assume that your circumstances mirror the general population's. If you have a different retirement age (earlier or later), different asset levels (have saved enough to fund your retirement fully with TIPS), or different needs for the money (for example, college savings) you need to consider what makes your situation different, and adjust your asset allocation accordingly.

Ability, willingness, and need

Main articles: Risk tolerance and Risk and return: an introduction

Risk is the uncertainty (variation) of an investment's return. Risk tolerance is your emotional and psychological ability to endure investment losses during large market declines without selling or undue worry, such as losing sleep. To know whether an asset allocation is right for your risk tolerance, you need to be brutally honest with yourself as you try to answer the question, "Will I sell during the next bear market?". This is very hard to accurately assess before you have already gone through a bear market.

Author Larry Swedroe has written a multi-part guide for selecting your asset allocation; how much to invest in stocks versus bonds.[note 5]

- **Ability:** Four factors determine your *ability* to take risk:[note 6][8]

1. Investment horizon - when do you need the money?
2. Stability of your earned income
3. The need for liquidity - if you need the money in a hurry
4. Your options if your existing plan fails to meet your objectives

- **Willingness:** Define your *willingness* to take risk.[9] Do you have the fortitude and discipline to stick with your predetermined investment strategy when the going gets rough?
- **Need:** The *need* to take risk is determined by the rate of return required to achieve financial objectives.[10] The greater the rate of return needed to achieve your financial objective, the more risks with equities you need to take. A critical part of the process is differentiating between real needs and desires. If you decide to take more risk because of perceived "need", keep in mind that taking extra risk could well backfire and lead to lower returns.
- How you should handle difficult choices among *ability*, *willingness*, and *need* to take risk.[11]

In general, Larry Swedroe suggests choosing the lowest equity allocation from among *ability*, *willingness*, and *need* to take risk (and adjusting goals as needed).[12]

In the case where the *ability* is greater than the *need* to take risk, it is possible to choose an equity allocation in the range between these two factors, based on your *willingness* to take risk.[12]

Rebalancing

Main article: Rebalancing

Over time your asset allocation may change from its original position as a result of the difference in returns from the various asset classes. Rebalancing is the act of bringing the asset allocation in line with current investment policy. A typical recommendation is that you should review the portfolio asset

allocation once a year, and if necessary, rebalance as specified in the investment policy. Rebalancing is often the most difficult part because it is counterintuitive, it requires you to sell a portion of an investment that went up, and buy more of what went down.[13]

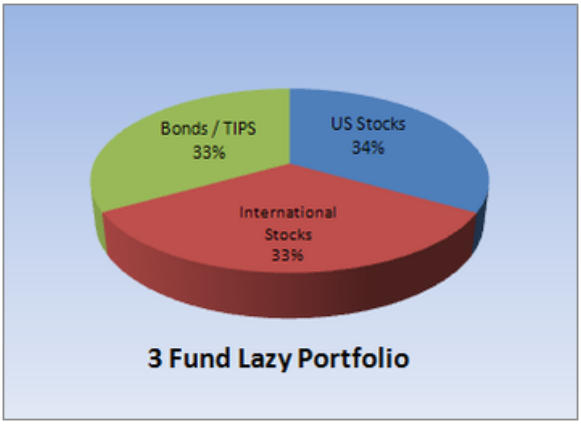
Asset allocation portfolios

Main article: Lazy portfolios

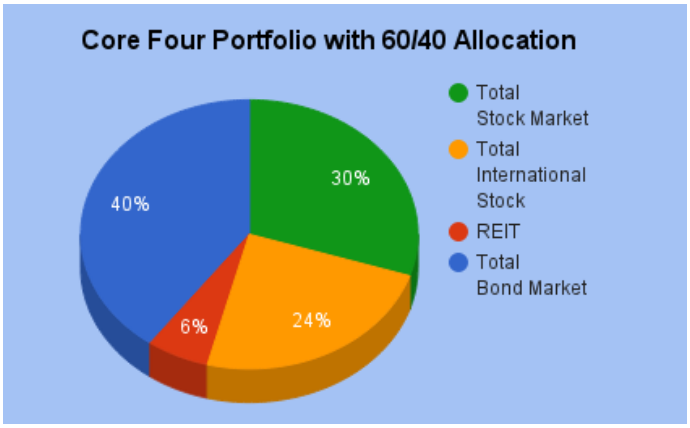
Strategic asset allocation strategies range from simple to complex. Lazy portfolios are designed to perform well in most market conditions. Most contain a small number of low-cost funds that are easy to rebalance. They are "lazy" in that you can maintain the same asset allocation for an extended period, suitable for most pre-retirement investors.

- John Bogle was a proponent of simple asset allocation portfolios. He frequently advised that most investors should allocate investment portfolios using two asset class index funds: a U.S. total market stock index fund, and a U.S. total bond market index fund.[14]
- A widely held portfolio among Bogleheads Forum members is the three fund portfolio. This allocates investments among a U.S. Total market stock market portfolio; a Total International stock market portfolio, and a U.S Total bond market portfolio. Members frequently expand this portfolio to include a fourth asset class, U.S. inflation-indexed bonds.
- Some strategic asset allocation funds add additional asset classes or sub-asset classes to the asset mix. For equity investments these additions can include value stock funds, real estate funds (U.S. and international), gold, and commodity futures funds. Additional fixed income asset classes include U.S high yield bond funds, international developed market bond funds, and emerging market bond funds. Depending on your risk tolerance preferences or tax situation, you might restrict your bond market allocations to U.S treasury bonds or investment grade municipal bonds.

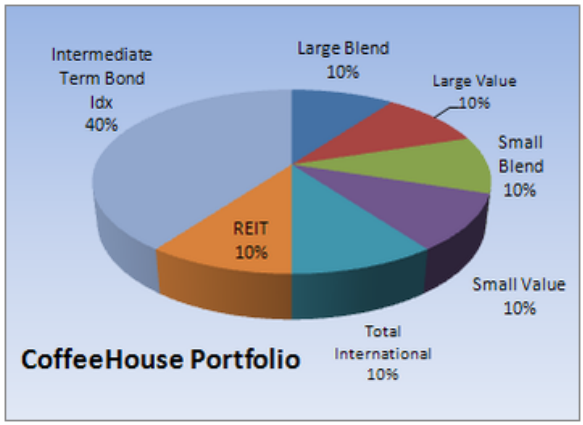
Sample strategic asset allocation portfolios



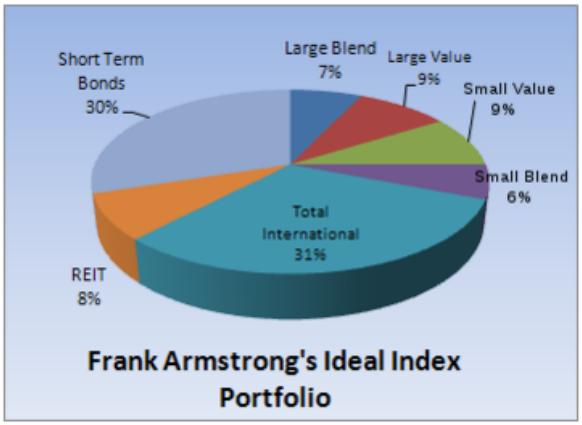
Three fund portfolio



Ferri Core four portfolio



Schultheis Coffeehouse portfolio



Armstrong Ideal Index portfolio

Funds that implement target asset allocations

Main articles: Balanced fund and Target date funds

An asset-allocation fund or a balanced fund is a mutual fund that holds multiple asset classes. Typically these funds hold stocks, bonds, and in some instances, cash. Although many balanced funds maintain a fixed asset allocation, some have a variable allocation policy, changing asset weightings according to market conditions.

Target date funds are balanced funds that gradually change asset class weightings according to a presumed changing need for a lower risk profile over time. These funds attempt to provide investors portfolios that account for the holder's age, risk appetite, and investment objectives, with an appropriate allocation of asset classes. However, critics of this approach point out that arriving at a standardized solution for allocating portfolio assets is problematic because individual investors require individual solutions.

Choices for equity

See also: Variations on Bogleheads® investing

For equity allocation, you need to choose the split between domestic ("US") and international (excluding US or "ex-US") stocks.

You may decide to select a set portion of regional stock markets, for example: North American markets, European markets, Asian markets, and Emerging markets.

If you are unsure of the US and ex-US allocation split (the sum representing the total amount of stocks), you can copy the allocations used by a balanced ("all-in-one") or target date fund. Those funds are created by experienced professionals who have done all the hard work before-hand. An example is shown in the wiki here.

At the same time, you need to decide whether to follow the full market or to select a portion (large-cap, mid-cap or small-cap); or to tilt in style (value, blend or growth); favor REITs, or to allocate among sectors.[note 7]

Choices for bonds

Bogleheads like to own bond funds instead of individual bonds for convenience and diversification. Using individual corporate or municipal bonds require a very large holding to achieve the broad diversification and increased safety of a bond fund. The high number of different bonds in bond funds let you ignore the risk of any one bond defaulting. You can manage Interest rate risk if you select funds with short and intermediate-term duration, and you can manage default risk by selecting funds with high credit ratings. The central idea here is that your bond holdings are for safety; that is, to reduce violent up and down swings in your overall portfolio value. Bogleheads tend to take risks on the equity side, not the bond side.

Bogleheads typically divide bond allocations between just two categories: nominal bonds such as the Vanguard Total Bond Market Fund (https://investor.vanguard.com/investment-products/mutual-funds/profile/vbtlx), and U.S. Treasury Inflation Protected Securities (TIPS) such as the Vanguard Inflation Protected Securities Fund (https://investor.vanguard.com/investment-products/mutual-funds/profile/vipsx). A TIPS fund gives you additional diversification, as well as inflation protection.[15]

I-Bonds are also an attractive alternative to TIPS. The U.S. Treasury sells these directly to investors. They do not need to be held in a tax-protected account; and they accrue interest tax-deferred for up to 30 years. There are annual limits on how much you can buy in I-bonds.

Notes

1. Investors use three asset allocation strategies.
   - **Strategic asset allocation** follows the Bogleheads investment philosophy. Strategic asset allocation is designed for the long run. The allocation may change due to life events, but it should not change due to market conditions.
   - **Tactical asset allocation**. Tactical asset allocation shifts allocations according to economic or valuation factors. Vanguard has historically used tactical asset allocation for a limited number of its balanced funds.
   - **Dynamic asset allocation**. Dynamic asset allocation calls for allocations to shift in accordance with changing future liabilities, a condition especially relevant, since changes in pension law, to institutional defined pension and institutional endowment portfolios, see The Pension Protection Act Of 2006 (https://web.archive.org/web/20160512131137/https://www.dol.gov/ebsa/pensionreform.html), Department of Labor, and Glide path ALM: A dynamic allocation approach to derisking (https://studylib.net/doc/8651971/glide-path-alm--a-dynamic-al

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2. Rule of thumb: A principle with broad application that is not intended to be strictly accurate or reliable for every situation. (Reference: Bogleheads forum post: "Wiki: Asset Allocation - Update "Age in Bonds"?" (https://www.bogleheads.org/forum/viewtopic.php?p=1445012#p1445012).)
3. Also see: Graham 75-25 rule
4. John C. Bogle elaborates his position, in the 2010 edition of *Common Sense on Mutual Funds*, pp.87-88:

> Long before the crash, I had fine-tuned my rule-of-thumb asset allocation model, centered at 50/50 for older investors in the distributions phase of their investment plan. Rather, I recommended — as a crude starting point — that an investor's bond position should be equal to his or her age. An investor age 65, then, would consider the propriety of a 65/35 bond/stock allocation. Clearly, such a rule must be adjusted to reflect an investor's objectives, risk tolerance, and overall financial position. (For example, pension and Social Security payments would be considered bondlike investments.) But the point is that as we age, we usually have (1) more wealth to protect, (2) less time to recoup severe losses, (3) greater need for income, and (4) perhaps an increased nervousness as markets jump around. All four of these factors suggest more bonds as we age.

"Age in bonds" and its variants, (age - 10) or (age - 20), are very general rules of thumb which you can adjust for your circumstances; a key circumstance being the presence or absence of a pension, which would change your willingness or need to take risk. Some Bogleheads do not add pensions and Social Security to their asset allocation of bond holdings. (Reference: Bogleheads forum post: "Wiki: Asset Allocation - Update "Age in Bonds"?" (https://www.bogleheads.org/forum/viewtopic.php?p=1445023#p1445023).)

5. Four articles by Author Larry Swedroe are referenced here:
   - **Part I:** Asset Allocation Guide: How much risk should you take? (http://www.cbsnews.com/news/asset-allocation-guide-how-much-risk-should-you-take/), CBS Moneywatch, February 3, 2014.
   - **Part II:** Asset Allocation Guide: What is your risk tolerance? (http://www.cbsnews.com/news/asset-allocation-guide-what-is-your-risk-tolerance/), CBS Moneywatch, February 12, 2014.
   - **Part III:** Asset Allocation Guide: How much risk do you need? (http://www.cbsnews.com/news/asset-allocation-guide-how-much-risk-do-you-need/), CBS Moneywatch, February 19, 2014.
   - **Part IV:** Asset Allocation Guide: Dealing with conflicting goals (http://www.cbsnews.com/news/asset-allocation-guide-dealing-with-conflicting-goals/), CBS Moneywatch, February 25, 2014.

   There are three additional articles in the series:
   - **Part V:** Asset Allocation guide: U.S. vs. international equity (http://www.cbsnews.com/news/asset-allocation-guide-us-vs-international-equity/), CBS Moneywatch, March 4, 2014. Determine the appropriate mix between domestic and international stocks. (This article is US focused.)
   - **Part VI:** Asset Allocation Guide: Value vs. growth (http://www.cbsnews.com/news/asset-allocation-guide-value-vs-growth/), CBS Moneywatch, March 13, 2014. Decide how much to invest

between *value* and *growth* stocks. (This article is intended for experienced investors who wish to deviate from the total market approach.)
- **Part VII:** Asset Allocation Guide: Small-cap vs. large-cap (http://www.cbsnews.com/news/asset-allocation-guide-small-cap-vs-large-cap/), CBS Moneywatch, March 18, 2014. Decide how much to invest between *small-cap* and *large-cap* stocks. (This article is intended for experienced investors who wish to deviate from the total market approach.)

6. *Ability* relates to your capacity to withstand the ups and downs of the market without getting nervous and making changes to the asset allocation. Selling in the face of a decline is about the worst thing you can do. Here is a table offered by author Larry Swedroe, based on the 1970s bear market, showing the amount of decline for various stock/bond allocations:

Asset allocation % (stock/bond)	Exposure to maximum loss	Asset allocation % (stock/bond)	Exposure to maximum loss
20/80	5%	70/30	30%
30/70	10%	80/20	35%
40/60	15%	90/10	40%
50/50	20%	100/0	50%
60/40	25%		

For example, an investor would be willing to accept a loss of 35% in the portfolio if they held an allocation of (80% stocks / 20% bonds). This table is from the 1970s; performance during other time periods will have different results. The general idea is for investors to select an asset allocation they are comfortable with. Source: Bogleheads forum topic: "Investment Planning" (https://www.bogleheads.org/forum/viewtopic.php?t=6211).

7. A common reasons for sector allocation is: You may be employed in a segment of the economy with low job security and so want to reduce or eliminate exposure to that sector in your investment portfolio.

See also

- Asset allocation in multiple accounts
- Asset classes (category)
- Asset allocation (category)
- Bogleheads® investing start-up kit
- Bogleheads investment philosophy
- Variations on Bogleheads® investing
- Lazy portfolios
- Tactical asset allocation

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2. John Bogle (2010). *Common Sense on Mutual Funds*. pp. 87–88.
3. "Strategic Asset Allocation - Definition of Strategic Asset Allocation on Investopedia" (https://www.investopedia.com/terms/s/strategicassetallocation.asp#axzz1mShM43qE). Investopedia. A portfolio strategy that involves periodically rebalancing the portfolio to maintain a long-term goal for asset allocation.
4. Joseph Davis; Daniel Piquet (2011). "Recessions and balanced portfolio returns" (https://web.archive.org/web/20210228023517/https://personal.vanguard.com/pdf/icrrbp.pdf) (PDF). Vanguard Institutional. Archived from the original (https://personal.vanguard.com/pdf/icrrbp.pdf) (PDF) on February 28, 2021.
5. Stock return from a Wilshire 5000 index fund; bond return from a Bloomberg Aggregate Bond Index fund; inflation data from US Treasury Department.
6. Input parameters use exactly the same return rate each year, with no variations, no variance, no co-variance. This is a simplified calculation for illustrative purpose only; actual returns will vary.
7. Benjamin Graham. *The Intelligent Investor* (2003 edition annotated by Jason Zweig ed.). Collins Business. p. 93. ISBN 978-0-06-174517-1.
8. Larry Swedroe (February 3, 2014). "Asset Allocation Guide: How much risk should you take?" (https://www.cbsnews.com/news/asset-allocation-guide-how-much-risk-should-you-take/). CBS Moneywatch.
9. Larry Swedroe (February 12, 2014). "Asset Allocation Guide: What is your risk tolerance?" (https://www.cbsnews.com/news/asset-allocation-guide-what-is-your-risk-tolerance/). CBS Moneywatch.
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13. Rick Ferri (2010). *All About Asset Allocation: The Easy Way to Get Started* (Second ed.). McGraw-Hill, Inc. p. x. ISBN 978-0071700788.
14. John Bogle argues this position in his classic text, *Common Sense on Mutual Funds*. ISBN 978-0470138137.
15. *The Bogleheads' Guide To Investing*. John Wiley & Sons, Inc. 2007. p. 103. ISBN 978-1-118-92128-9.

Further reading

Importance of asset allocation

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External links

- Bogleheads forum topic: "Wiki comments requested: The importance of asset allocation" (https://www.bogleheads.org/forum/viewtopic.php?t=156946)
- Bogleheads forum topic: "Wiki: Asset Allocation - Update "Age in Bonds"?" (https://www.bogleheads.org/forum/viewtopic.php?t=99826)
- Google search (https://www.google.ccom/search?q=define:Asset+Allocation) for definitions of 'Asset Allocation'

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